Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*2989647625*

**MATHEMATICS (SYLLABUS D)** **4024/12**

Paper 1 **May/June 2020**

**2 hours**

You must answer on the question paper.

You will need: Geometrical instruments

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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**[Turn over**

**ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER**

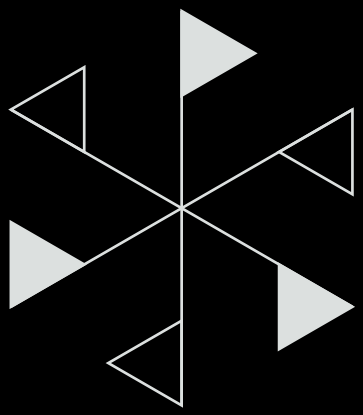
**1** **(a)** Work out $0.05 \times 0.3$.

..... [1]

**(b)** Work out $2\frac{2}{3} - \frac{1}{5}$.

..... [2]

**2** **(a)**



Write down the order of rotational symmetry of this shape.

..... [1]

**(b)** Samuel describes a special quadrilateral.

It has only one line of symmetry.
Its diagonals cross at right angles.

Write down the name of this special quadrilateral.

..... [1]

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**3** Write these numbers in order of size, starting with the smallest.

$4^3$ $9^2$ $\sqrt{196}$ $\sqrt[3]{125}$

..... , ..... , ..... , ..... [2]
*smallest*

**4** **(a)** Write 68% as a fraction in its lowest terms.

..... [1]

**(b)** A bag contains red balls and blue balls.
The balls are in the ratio red : blue = 3 : 5.

Write down the fraction of the balls that are red.

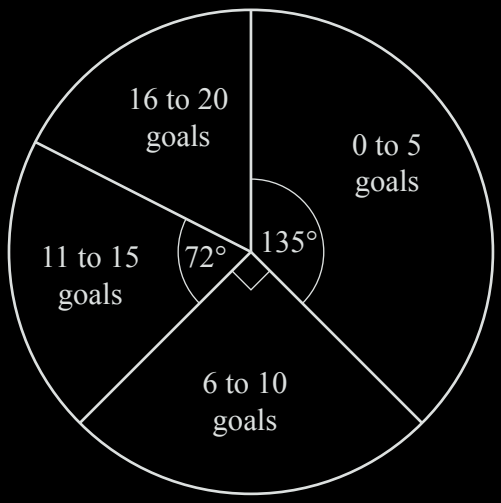
..... [1]

**5** By writing each number correct to one significant figure, estimate the value of

$$\frac{2.78^3}{61.4 \times 0.893}.$$

..... [2]

**6**




The pie chart shows information about the number of goals scored by each player in a football club.

**(a)** Write down the modal class.

..... [1]

**(b)** 8 of the players each scored 11 to 15 goals.

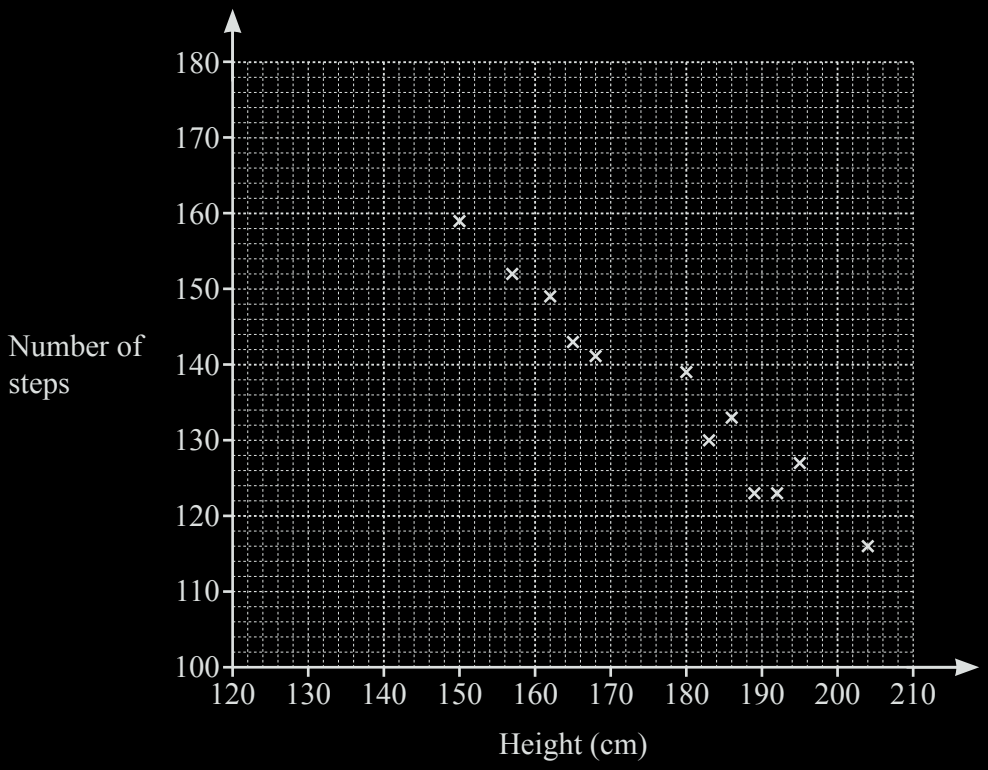
Work out the total number of players in the club.

..... [2]

**7** Factorise $15a - 5x - 2xy + 6ay$.

..... [2]

**8** The number of steps taken by 12 people to walk 100 m was recorded.
The scatter diagram shows the heights of these people and the number of steps they took.




**(a)** What type of correlation is shown in the scatter diagram?

..... [1]

**(b)** Draw a line of best fit. [1]

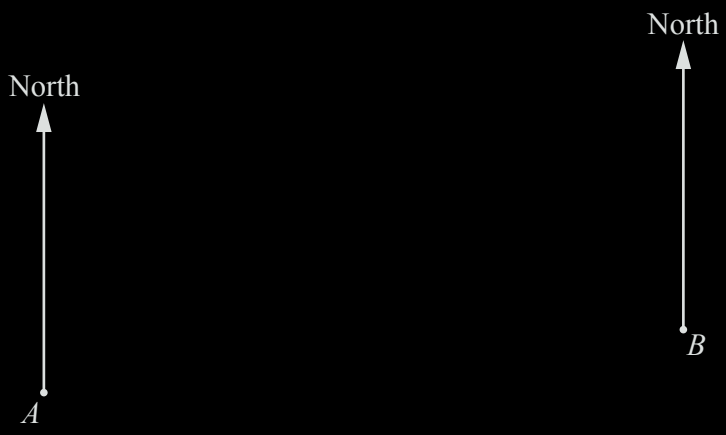
**(c)** The height of another person is 175 cm.

Use your line of best fit to estimate the number of steps they would take to walk 100 m.

..... [1]

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**9**




**Scale: 1 cm to 10 km**

The scale drawing shows the positions of town *A* and town *B*.

**(a)** Find the actual distance, in kilometres, of town *A* from town *B*.

..... km [1]

**(b)** Town *C* is on a bearing of 140° from town *A* and on a bearing of 235° from town *B*.

Mark the position of town *C* on the scale drawing. [2]

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**10** **(a)** Bilal goes for a cycle ride.
He starts at 3 pm.
He finishes at 5.38 pm.
He has a total of 25 minutes rest during the ride.

Work out how long, in hours and minutes, he spends cycling.

..... hours ..... minutes [1]

**(b)** Sonia walks to her aunt's house.
She leaves home at 10 25.
She walks a total of 12 km at an average speed of 5 km/h.

Work out the time Sonia arrives at her aunt's house.

..... [3]

**11** **(a)** $c = \frac{7-a}{b}$

Find $c$ when $a = -4$ and $b = 2$.

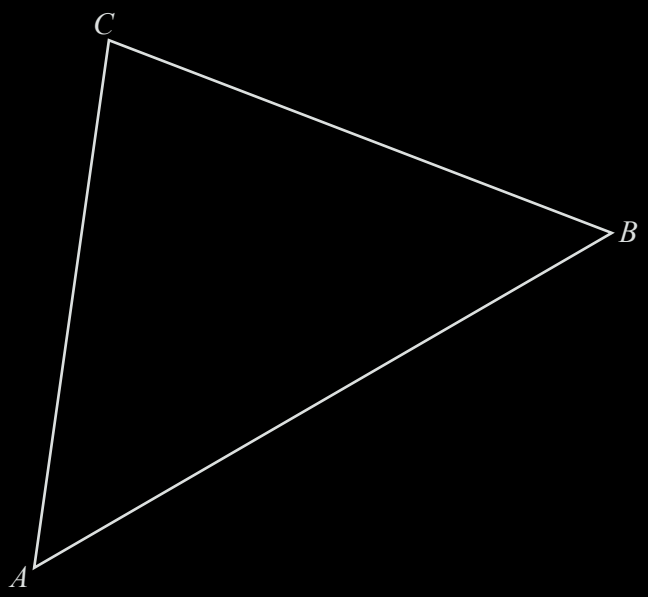
$c =$ ..... [1]

**(b)** $y = 5^x + 1$

Find $y$ when $x = -2$.

$y =$ ..... [1]

**12 Use a straight edge and compasses only in this question.**




**(a)** Construct the locus of points inside triangle $ABC$ that are

**(i)** 5 cm from $B$, [1]

**(ii)** equidistant from $A$ and $C$. [2]

**(b)** Shade the region inside triangle $ABC$ containing the points that are

- less than 5 cm from $B$

and

- closer to $A$ than to $C$. [1]

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**13** **(a)** Write 108 as the product of its prime factors.

..... [2]

**(b)** Find the lowest common multiple (LCM) of 108 and 180.

..... [2]

**14** **(a)** In 2017, the population of Egypt was 97 500 000.

Write this population in standard form.

..... [1]

**(b)** The population density of a country is the number of people per square kilometre.

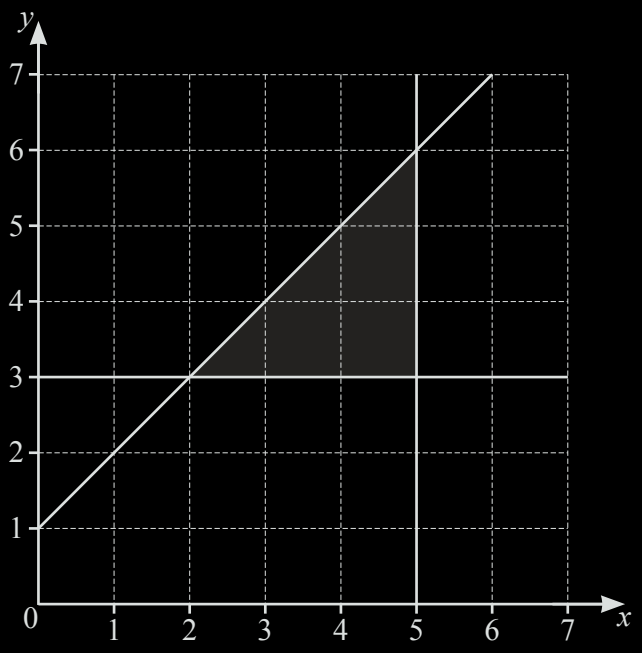
In 2017, the population of Indonesia was $2.62 \times 10^8$, correct to 3 significant figures.
The area of Indonesia is $2 \times 10^6 \text{ km}^2$, correct to 1 significant figure.

Calculate an estimate for the population density of Indonesia.

..... people/$\text{km}^2$ [2]

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**15**




The shaded region is defined by three inequalities.

Find these three inequalities.

.....

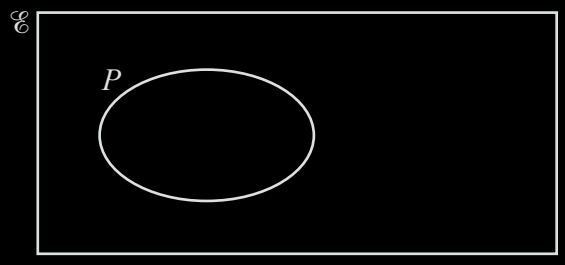
.....

..... [3]

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**16** $Q \subset P$
$P \cap R = \varnothing$

Complete the Venn diagram to show sets $Q$ and $R$.




[2]

**17** Here are the first four terms of a number sequence.

$T_1 = 1^2 + 3 \quad = 4$

$T_2 = 2^2 + 8 \quad = 12$

$T_3 = 3^2 + 13 = 22$

$T_4 = 4^2 + 18 = 34$

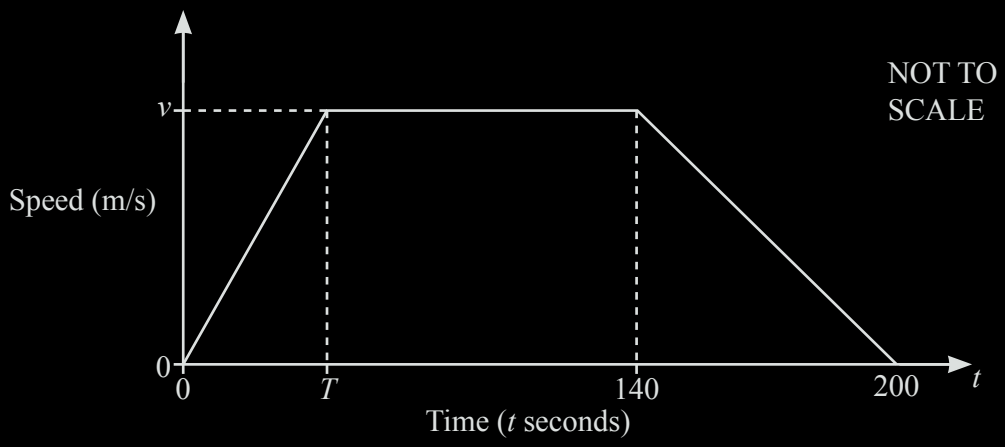
**(a)** Find $T_5$.

$T_5$ = ..... [1]

**(b)** Find an expression, in terms of $n$, for $T_n$.

$T_n$ = ..... [3]

**18** The diagram is the speed–time graph for part of a car's journey.




**(a)** The deceleration of the car between $t = 140$ and $t = 200$ is $0.2\,\text{m/s}^2$.

Find the value of $v$.

$v =$ ..... [2]

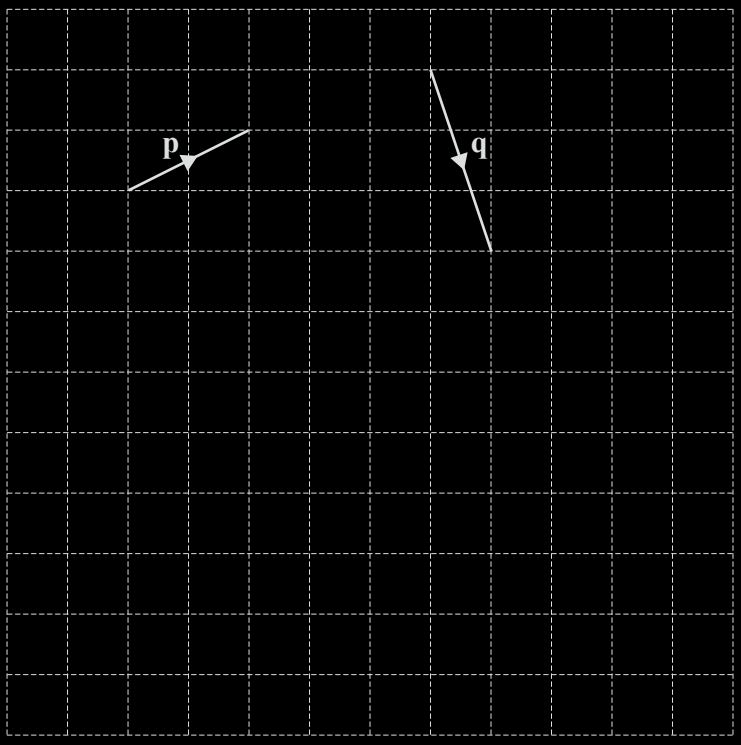
**(b)** The car travels a total of 1800 m in the 200 seconds.

Find the value of $T$.

$T =$ ..... [3]

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**19**




Vectors **p** and **q** are shown on the grid.

On the grid, draw the vector

**(a)** $3\mathbf{p}$, [1]

**(b)** $\mathbf{q} - \mathbf{p}$. [1]

**20** A plan of a house is drawn to a scale of 1 : 50.
On the plan, the floor area of the kitchen is $30\,\text{cm}^2$.

Calculate the floor area of the real kitchen.
Give your answer in square metres.

..... $\text{m}^2$ [3]

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**21** Simplify $\left(\frac{2x^2}{x^5}\right)^{-3}$.

..... [2]

**22** $f(x) = 4(3-x)$ $g(x) = \frac{5(3x-2)}{x}$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... [2]

**(b)** Solve $g(x) = 6$.

$x =$ ..... [3]

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**23** Express as a single fraction in its simplest form.

$$\frac{5}{2x-1} - \frac{3}{x+4}$$

..... [3]

**24** $P$ is the point $(h, 7)$.
$P$ lies on the line $3y + 2x = 5$.

**(a)** Find the value of $h$.

$h =$ ..... [2]

**(b)** Line $L$ is perpendicular to the line $3y + 2x = 5$ and passes through $P$.

Find the equation of line $L$.

..... [4]

**Question 25 is printed on the next page**

**25** $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ -3 & -1 \end{pmatrix}$

**(a)** Evaluate $2\mathbf{A} - \begin{pmatrix} -5 & 4 \\ 0 & 3 \end{pmatrix}$.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

**(b)** Find $|\mathbf{A}|$.

..... [1]

**(c)** Find $\mathbf{A}^{-1}$.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [1]

**(d)** Find the matrix $\mathbf{X}$, where $\mathbf{XA} = \begin{pmatrix} 4 & -2 \end{pmatrix}$.

$\mathbf{X} =$ [2]

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